



In situ X-ray imaging and quantitative analysis of balling during laser powder bed fusion of 316L at high layer thickness

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ABSTRACT

Balling is the main surface defect in additive manufacturing, leading to surface roughness and uneven powder deposition. Through the in-situ X-ray imaging technology, the melting process of high layer thickness 316L powder under different process parameters was investigated in real time in this work. We systematically elaborate the complex formation mechanism of balling at high layer thickness, and the key mechanism underlies the splatter's coalescence during the flight and solidification stages. The frequent splatters coalescence dominates the large-size balling. The spatter coalescence event was roughly quantified, and the coalescence rate ranges from 42.42 % to 73.04 %. The swing of the irregular balls and jumping of the regular small balls were observed, and the solidification time ranges from 10 ms to 20 ms. Moreover, the detailed morphological parameters including the contact angle and counts of the spatter were clarified, and the algebraic equations about the contact angle and the volumetric energy density were established. This study provides a systematical understanding of the balling phenomenon during laser powder bed fusion of 316L at high layer thickness.

1. Introduction

Laser powder bed fusion (L-PBF) additive manufacturing (AM) enables the fabrication of parts with precise dimensional control, freedom of design, and material properties [1–4], which is a potentially disruptive technology across multiple industries, including aircraft, aerospace, biomedical and automotive [5–9]. LPBF usually using thin layer thicknesses of 20–60 μm to ensure the fabrication of metal parts with a finer surface roughness and a higher relative density. However, the thin layer thickness greatly limits its building efficiency [10–13], for example, Thijs et al. [12] and Song et al. [13] separately used 30 μm and 50 μm layer thickness to fabricate Ti-6Al-4V samples, the building efficiency of which was 0.45 mm^3/s and 0.8 mm^3/s respectively.

Recent researches demonstrate that using high layer thickness can improve the building efficiency because of the reduction of the powder laying times and the subsequent required fabrication time [14–16]. Such as, Wang et al. [15] increased the layer thickness to 150 μm to produce 316L alloy parts, deriving a building rate up to 12 mm^3/s , which is about 3–10 times higher than the traditional LPBF processing. None-theless, it is worth noting that, according to many studies, when the layer

thickness exceeds 100 μm , it is liable to cause generation of balling and spattering, resulting in defects and rough surface [10,17–19].

The balling [20] phenomenon describes the process of continuous molten pool splitting into isolated melt (also known as “humps” [21,22]). Large-size balling has significant adverse effects on industrial printing, such as non-uniform powder distribution in the consecutive powder layer, poor wettability of the previous layer [23,24], and even destroying the build process due to a collision with the paving roller [25]. The final parts show a balling porosity [26], rough surface [25], or poor inter-layer bonding [27]. Therefore, a thorough understanding of the formation process and mechanisms of the balling is required for a reliable fabrication process.

Numerical simulation can reproduce the balling phenomenon and allow for analysis of the melt pool breakups. Zoller et al. [28] have used the smoothed particle hydrodynamics (SPH) with accurate interface modeling to derive balling as the formation of isolated balls by contraction of the melt pool tail. Liu et al. [8] and Shi et al. [29] found that balling at large layer thicknesses resulted in the agglomeration of some incompletely melted powders. Though the numerical simulation has been an important tool for studying the balling formation, it cannot

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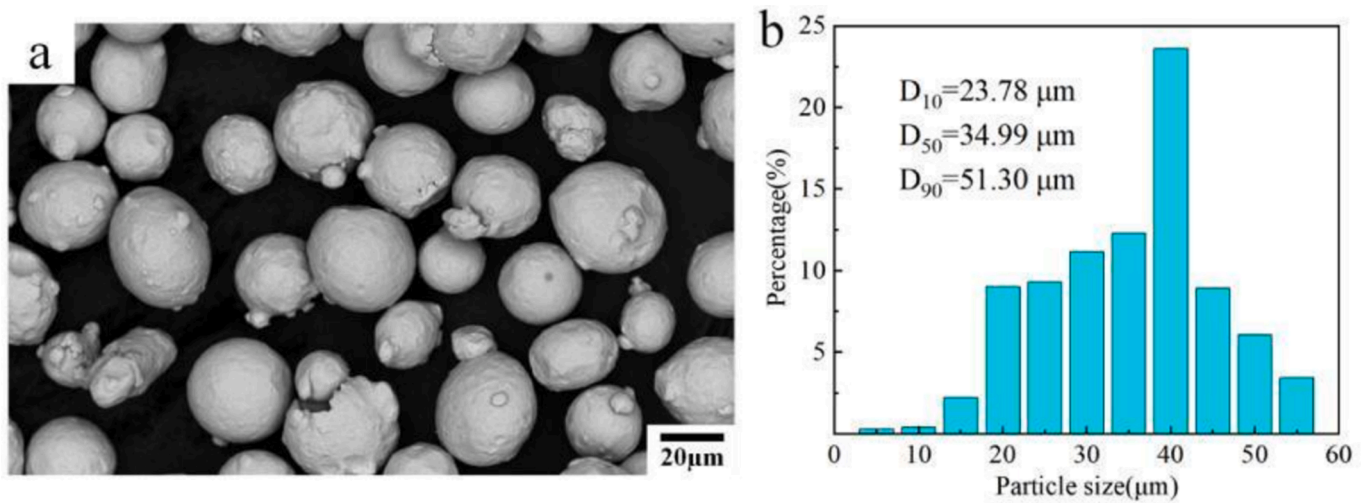


Fig. 1. Morphology and size distribution of 316L stainless steel powder.

well reproduce the multiple complex physical phenomena (e.g. spatter generation and coalescence [30]) involved in LPBF, leading to the information loss about the actual balling formation. The balling can also be observed at both the high or low laser energy density [31–33]. Several mechanisms for balling have been proposed. For some materials, surface tension decreases with increasing temperature, and the severely unstable Marangoni flow causes the melt pool breakup, forming a balling phenomenon [34]. The balling occurs due to the relatively higher capillary forces when poor wettability exists between the melt and its surrounding powder or previous melt layer [27,35–37]. The balling also relates to the partially melted powder or insufficient remelting of the previous layer due to the small amount of the melt [38]. In summary, insufficient output energy is one of the contributing factors to the occurrence of the balling phenomenon.

In addition, the balling phenomenon and the spattering phenomenon are closely related during the laser powder bed fusion process. Spattering is used to describe the material ejection from melt pools during LPBF printing process [39]. This phenomenon can promote balling to some extent and increase the surface roughness of the workpiece [40]. Gu et al. [41] found that balls formation was ascribed to laser-induced melt splashes caused by a high capillary instability of the melt. Simionelli et al. [42] found that the spatters mainly contribute to the generation of the small balls on the substrate or melt track surface. It can be found that the formation process and mechanism of the balling at high layer thickness are complicated, and the real-time observing of the balling formation process by the in-situ method is quite important to further reveal its mechanism.

This work aims to close the gap between the balling and the spatter using the in-situ micro-focus X-ray imaging system in laser melting process at high layer thickness of 316L. We focus on the contribution of the spatters' coalescence on the balling formation monitoring the laser-powder interaction at different process parameters. Besides, we quantified the spatters' coalescence events and the morphological characteristics of the balls. This work provides an intuitive and deep understanding of the balling mechanism at high layer thickness LPBF of 316L.

2. Materials and methods

2.1. Power materials

The 316L powder was used for a single melt track in this work, and the powder was prepared by the gas atomization method (AVIC Mate Additive Technology (Beijing) Co., Ltd). The powder had a spherical shape (Fig. 1(a)), and the D₁₀, D₅₀, and D₉₀ are 23.78 μm, 34.99 μm and 51.30 μm (Fig. 1(b)), which means the particle size of the powder at 10 %, 50 %, and 90 % cumulative distribution respectively. The chemical composition of 316L powder is shown in Table 1.

2.2. In-situ observation system

The micro-focus X-ray imaging system for observing the LPBF process of 316L is shown in Fig. 2. The working principle bases that the X-ray can image due to the density difference between the Argon, accumulated powder, and the substrate, then the X-ray images are converted into the visible images. A micro-focus X-ray machine (Swiss FXE 225.48, 225 kV, 1.4 mA, with a spatial resolution of 3 μm) was utilized. The system also includes an image intensifier (Japan CANON ELECTRON TUBES & DEVICES CO, with a resolution of 70/110 Lp/cm and an output image diameter of 20 mm), a high-speed camera (FASTCAM Mini AX200 series, Tianjin Jie Xin Technology Co., Ltd., with a resolution of 1024 × 1024, 6400 fps at full frame, and a minimum exposure time of 1.05 μs), and micro 3D-printer (Wuhan Ruikang Fiber Laser Co., Ltd., with maximum laser power of 450 W, a wavelength of 1080 nm, and a spot diameter of 200 μm). The system was designed to meet the latest international radiation safety standards.

The powder bed samples consist of a layer of powder on top of a substrate, sandwiched together between two glassy carbon plates. The horizontal thickness of the 316L substrate is 0.55 mm, determining the horizontal powder bed thickness of 0.55 mm. The layer thickness of the powder bed is determined by the height difference between the substrate and the glassy carbon plates. In each experiment, the printer chamber was pumped down first and then purged with pure Ar gas. Before the laser scanning process, we calibrate the position of the laser focus using a CCD camera to ensure the laser path aligns with the centerline of the powder bed, thereby avoiding contact between the melt

Table 1
Chemical composition of 316L powder.

Composition	Fe	Cr	Ni	Mo	Mn	Si	P	S	C
Wt. %	Bal	17.2	10.8	2.45	0.72	0.44	0.01	0.02	0.08

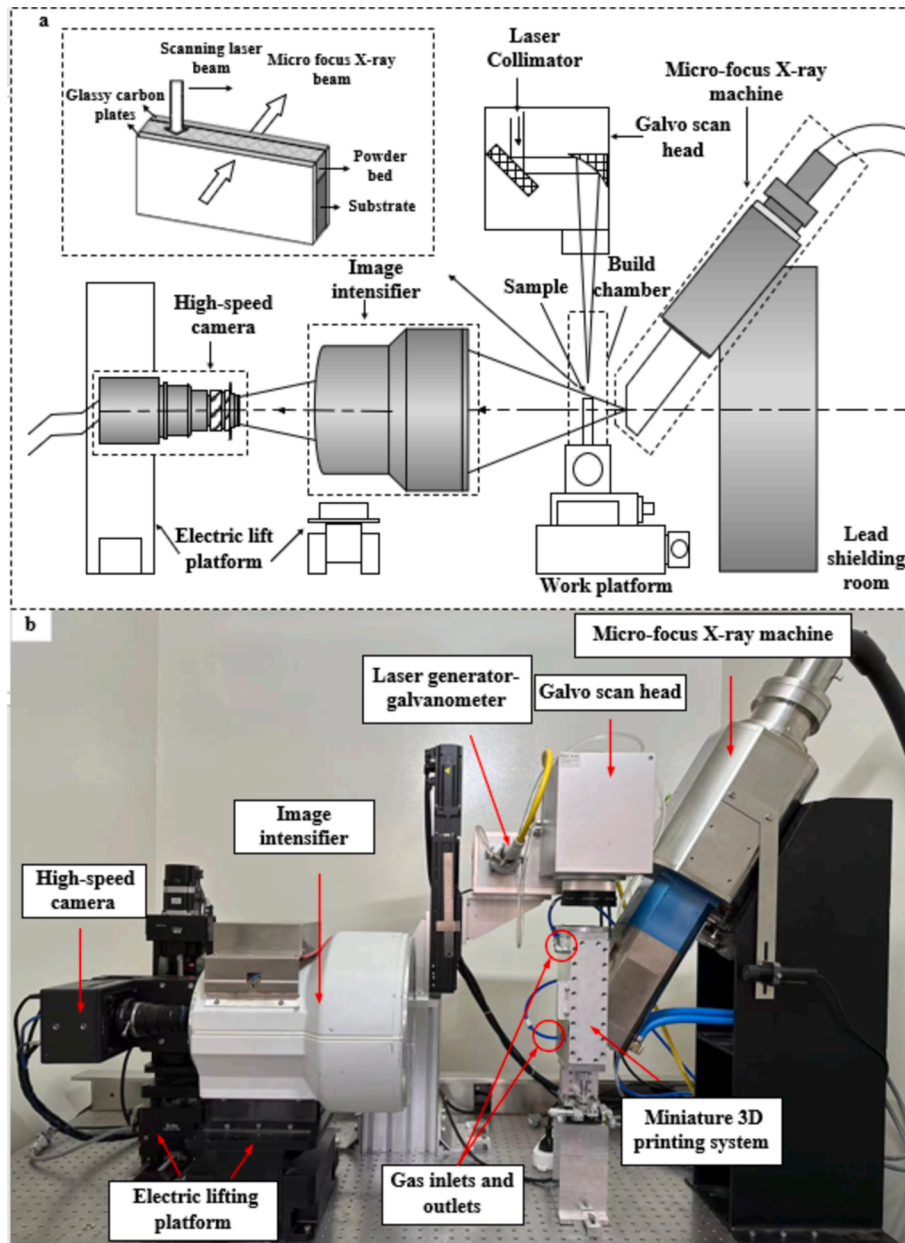


Fig. 2. In-situ X-ray imaging system schematic diagram and experimental device.

Table 2

Process parameters for LPBF of 316L.

LT (μm)	P (W)	V (mm/s)
200	45, 90, 180, 270, 360, 450	30, 60, 90, 120, 150
100	45, 90, 180, 270	60
400	45, 90, 180, 270	60

pool and the glass plate.

2.3. Parameters for the LPBF process

The series of single tracks of 316L were fabricated by varying process parameters such as laser power (P), scanning speed (V), and layer thickness (LT), as given in Table 2.

The powder layer thickness usually selected is 30–60 μm in the industrial printing process [43,44]. The in-suit experiments were conducted to explore the balling process at the high layer thicknesses of 100

μm , 200 μm , and 400 μm in this work. But note that the large powder bed layer thickness causes low scanning speeds slower than commonly used industrial conditions at the same laser power.

The thickness of 200 μm was selected as a starting point for the in-situ study, and the complete melting processes including balling and continuous forming were first observed at different laser powers and scanning speeds. The results at the scanning speed of 60 mm/s were included in Section 3.1.1 as they exemplify the types of features present. Then we focus on the process parameters of laser power and scanning speed to explore the balling phenomenon at different layer thicknesses of 100 μm and 400 μm .

The effects of volume energy density on the contact angle and count of the balls were investigated. The calculation formula of volume energy density:

$$Ev = \frac{P}{V \cdot d \cdot LT} \quad (1)$$

where Ev is the volume energy density, J/mm^3 ; P is the laser power, W; V

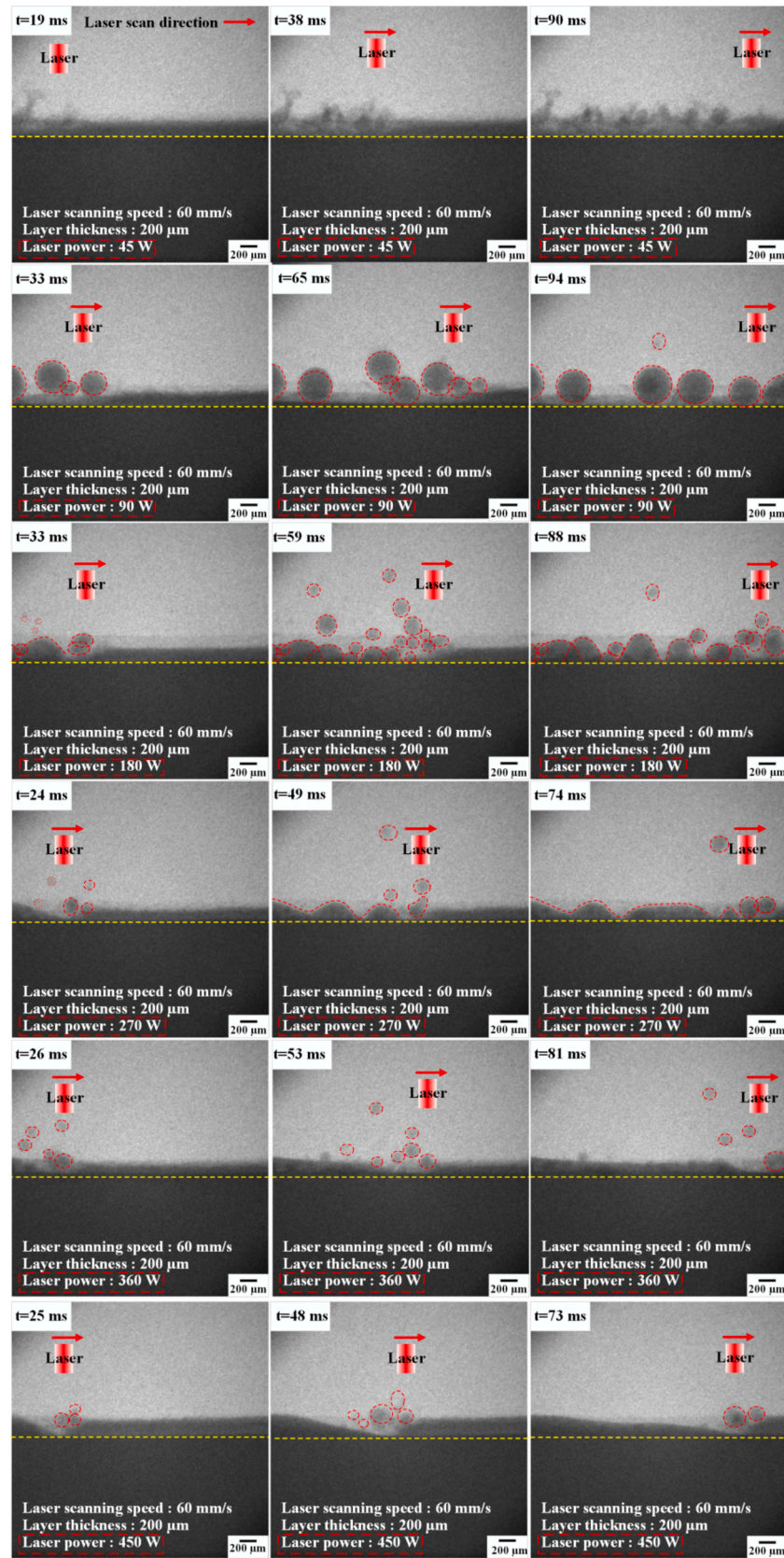


Fig. 3. Powder melting process at different laser powers (The yellow line represents the boundary between the powder bed and the substrate).

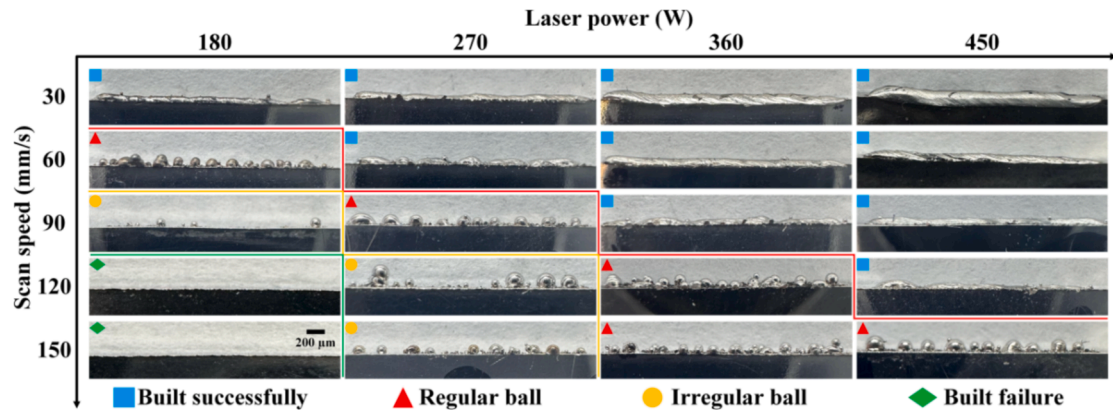


Fig. 4. Morphology of melt track at different process parameters ($LT = 200 \mu\text{m}$).

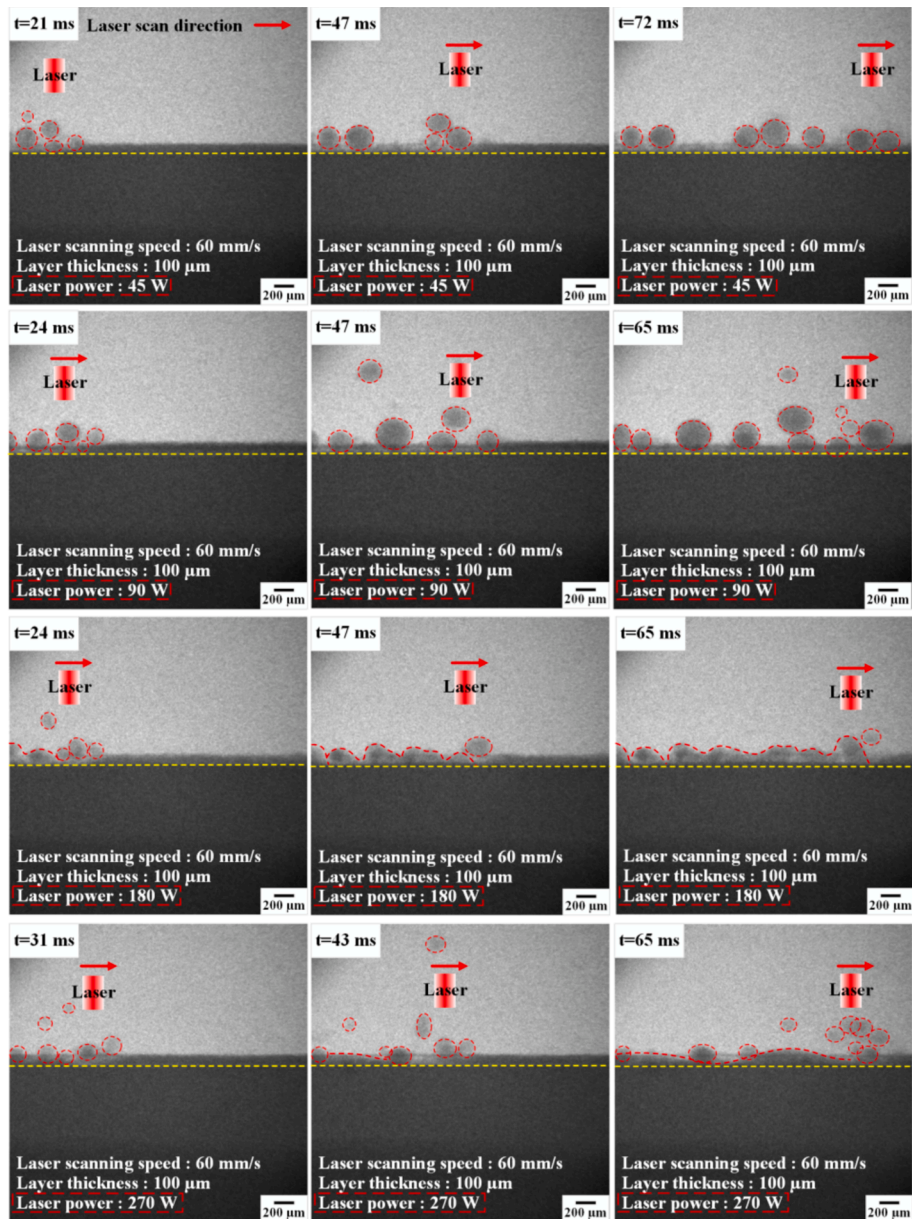


Fig. 5. Powder melting process at layer thickness of 100 μm.

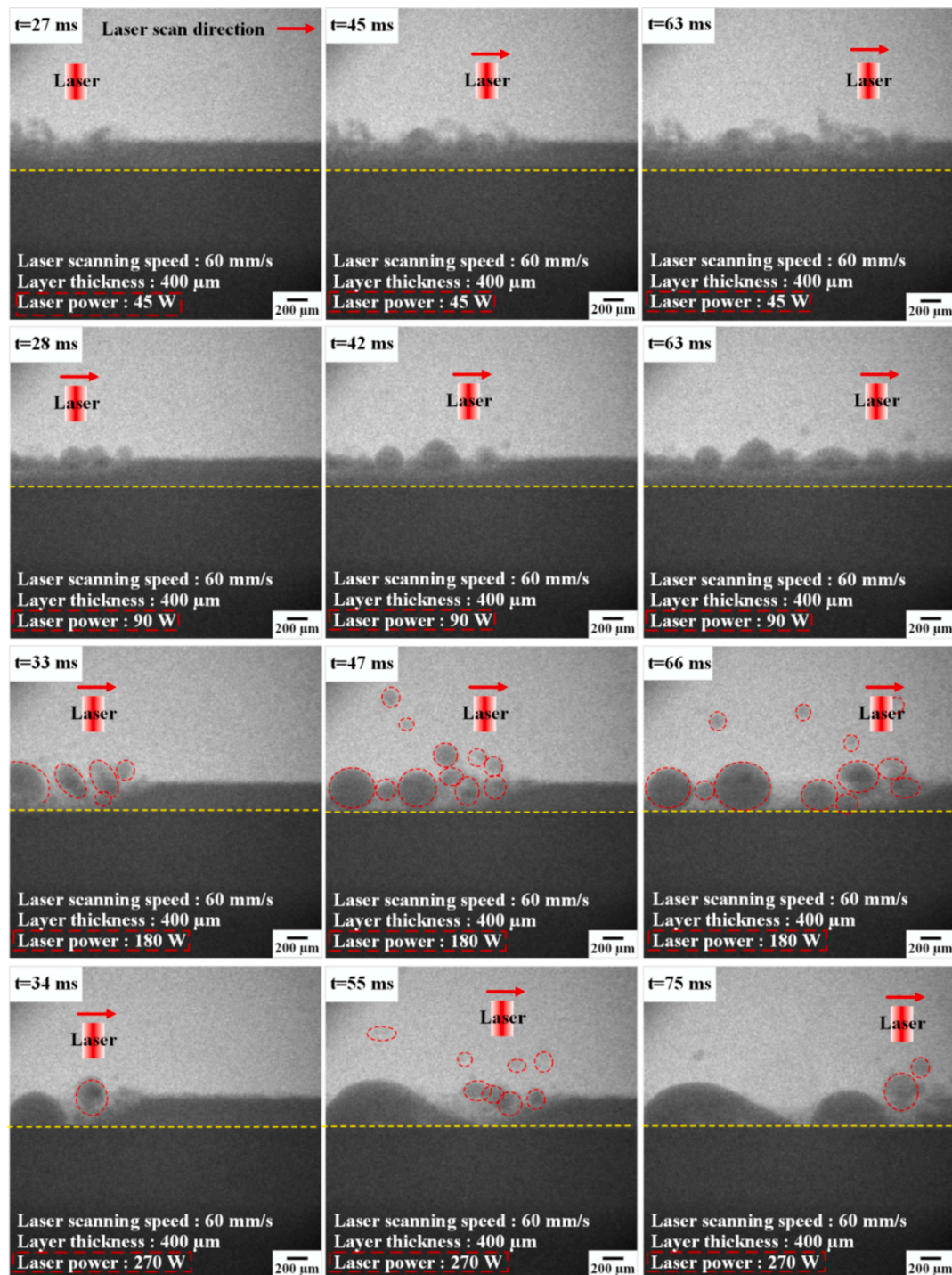


Fig. 6. Powder melting process at layer thickness of 400 μm .

is the scanning speed, mm/s; d is the spot diameter, μm ; LT is the powder layer thickness, μm .

3. Results and discussion

3.1. Formation process of balling

3.1.1. Influence of the laser power

The real-time melting processes of the 316L powder with a layer thickness of 200 μm at different laser powers and scanning speeds were given in the [Movie S1–S5](#). The representative morphologies are shown in

[Fig. 3](#) (see [Movie S2](#)). The laser fails to penetrate the powder bed because of the insufficient laser energy at $P = 45$ W, and the mixture of the molten droplets and powder is formed on the substrate. Spatters caused by the recoil pressure are mainly the cold spatter under such conditions, and the spatters' coalescence rarely happens. With increasing the laser power to 90 W, the laser just penetrates the powder bed. Some powder melt rapidly and then shrinks, forming small balls. Note that the spatters coalesce frequently during the flight and solidification, leading to the large-size balls. In addition, the jumping phenomenon of the balls with small sizes was observed. It should be noted that the balls can fall off the substrate at $P = 45$ W and 90 W. With

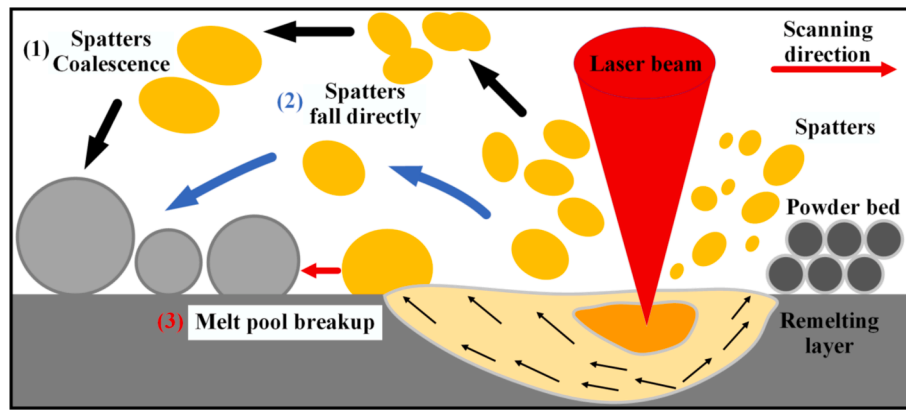


Fig. 7. Schematic diagram of the balling formation process at high layer thickness.

increasing the laser power to 180 W, the laser can completely melt the powder, and more smaller hot spatters are ejected. The active hot spatters coalesce with each other, forming the irregular-shaped large-size balls due to the improved wettability between the metal liquid and the substrate.

When the laser power exceeds 270 W, the melt pool deepens and widens, and the count and flight height of the spatter decreases instead. The melted substrate offers sufficient metal liquid, and the ball count is significantly reduced, tending to generate a continuous melt track.

The morphology of the melt track at different process parameters is shown in Fig. 4. For the given laser scanning speed, the powder melting process develops from built failure, irregular balling, regular balling, and continuous melt track with increasing the laser power [20]. The

green area represents that no melt track can be generated due to the very low laser power and not completely melted powder. In the yellow area, the balls are featured with small spacing and low counts, and large differences in size due to the poor wettability between the melt pool and its surrounding powder and substrate [35,45,46], as well as the spatter's coalescence. Further increasing the laser power (red area), the melt pool is torn into the uniformly distributed melt, and the spatters coalescence promotes the growth of the melt, causing the regular large-size balls with similar dimensions. The main reasons for the uneven distribution of the balls lay in the randomness of the spatter drop point and the melt pool broken. Larger laser power at the red area leads to sufficient melting of the substrate, which eliminates the balling phenomenon [47].

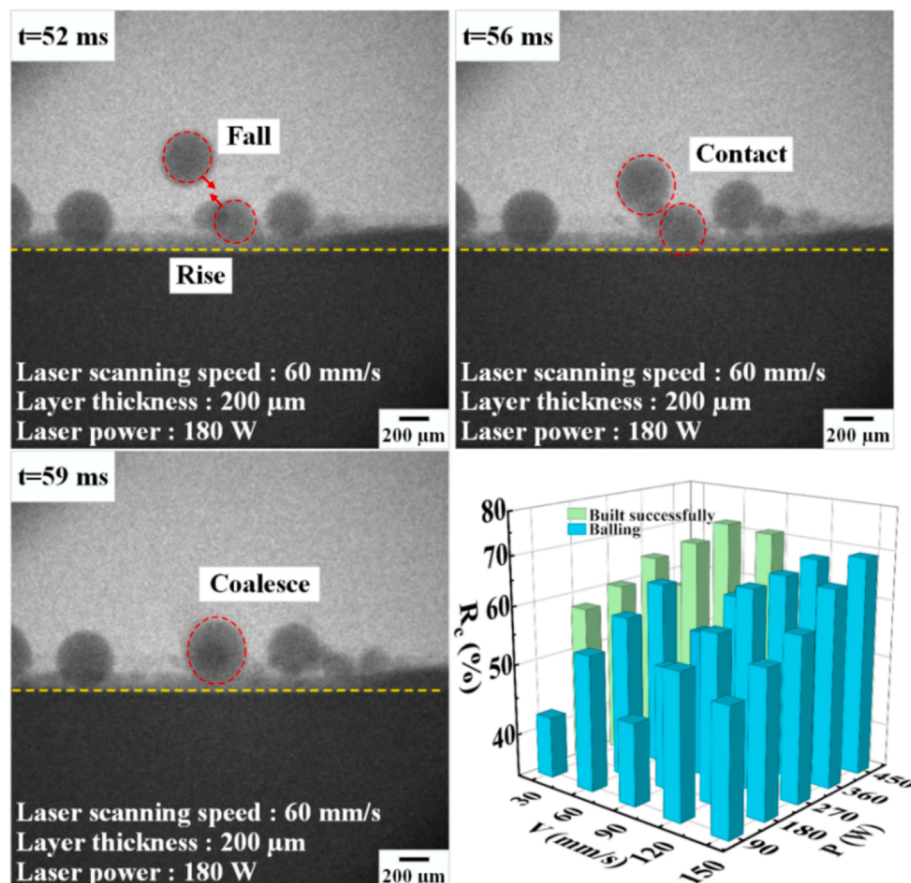


Fig. 8. The process of the spatters' coalescence and variation of the spatters' coalescence rate.

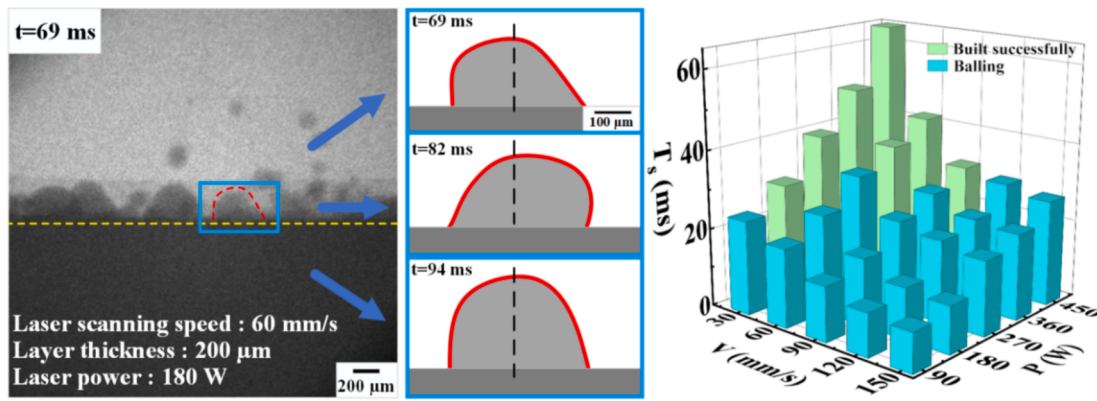


Fig. 9. The process of the melt solidification and variation of the solidification time.

3.1.2. Influence of powder bed layer thickness

Powder bed thickness is a key process parameter in determining the melt pool and single tracks [48,49]. As expected, the laser can easily penetrate the powder and substrate because of the low powder bed thickness of 100 μm at $P = 45$ W and 90 W, and the regular balls are distributed on the substrate, as shown in Fig. 5 (see Movie S6). Similarly, we observed several spatters coalesce, leading to the large-size balls. With further increasing the laser power, the balling phenomenon disappears due to the sufficient metal liquid although the spatters coalescence still exists.

When increasing the powder bed layer thickness to 200 μm , a large amount of powder cannot be melted due to the insufficient laser energy at $P = 45$ W and 90 W, as shown in Fig. 6 (see Movie S7). The melt pool is small and submerges into the thick powder bed, which causes a small number of spatters. No melt track is formed on the substrate and the spatters are mainly cold spatters in this condition. At $P = 180$ W, the spatters are mainly hot ones, and there appears obvious balling phenomenon and spatters coalescence phenomenon. The balls become regular in shape and bigger in size. With increasing the laser power to 270 W, the melt pool is firstly fractured into discontinuous irregular melts due to the improved wettability. Then the spatters constantly merger with those melts, leading to the irregular large-size balls. The metal liquid is completely spread, generating a continuous melt track at higher laser powers of 360 W and 450 W. The above results indicate that a larger layer thickness requires larger laser energy to avoid the balling, and the large-size balls are generated mainly due to the spatters' coalescence events.

3.1.3. Formation mechanism of balling at high layer thickness

After summarizing the results of in-situ experiments for the powder melting process under different process parameters above, the balling formation under large layer thickness mainly involves three processes, as shown in Fig. 7. (1) Spattering occurs during the LPBF process due to the vapor recoil pressure and Marangoni effect [6], or vapor-induced entrainment [7,8]. During the flight, some spatters contact and coalesce into big spatters, and then fall on the substrate, forming the unsolidified large-size balls. During the solidification, some spatters may

continue to merge into the unsolidified balls, contributing to the large-size balling. (2) Some spatters directly fall on the substrate and often form small balls. (3) The metal liquid at the end of the melt pool breaks, contracts, and cools, which dominates the formation of the medium balls. Note that these three processes are not independent. For instance, some spatters may merge into the broken metal liquid, which also causes the medium balls to grow into large-size balls. The process (1) dominates the balling formation at low laser energy. With the increase of laser energy, more discontinuous melts are generated by the process (3). Process (2) has always existed during the powder melting, but the volume proportion of the small balls is relatively low.

3.2. Spatter coalescence rate and ball solidification time

As mentioned above, the spatter's coalescence is an important factor affecting the balling phenomenon at high layer thickness. Fig. 8 shows the representative process of the spatter coalescence event. The powder melts under the action of a laser, which ejects two large-size hot spatters with diameters of about 285.7 μm and 257.1 μm at $t = 52$ ms, respectively. One spatter at falling contacts with the other spatter at rising at $t = 56$ ms, and then the spatter coalescence occurs, forming a large-size ball with a diameter of about 342.8 μm . Note that more than two spatters coalesce sometimes, and the spatters may merge into unsolidified balls.

The coalescence rate (R_c) was introduced to quantify the spatters coalescence event, which presents the proportion of the coalesced spatters count to the total captured spatters count. The statistical result shows that the laser power has a more significant effect on the spatter coalescence event. Higher laser power leads to more spatters coalescence, and the spatters coalescence rate ranges between 42.42 % and 73.04 %.

The solidification time (T_s) is an index that signs the solidification process of the balls that adhered to the substrate. The surface fluctuation of the ball can be observed, as shown in Fig. 9 (see Movie S8). The ball begins to swing at $t = 69$ ms, and the ball solidifies marked by stopping swinging at $t = 94$ ms. Then the solidification time of the balls can be approximately obtained. Because of the smaller laser power and faster

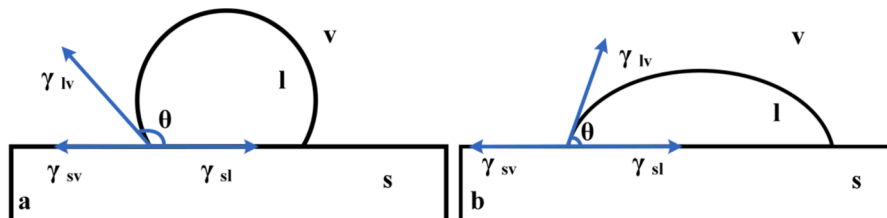


Fig. 10. Schematic diagram of contact angle: a. $\theta > 90^\circ$, bad wettability. b. $\theta < 90^\circ$, perfect wettability.

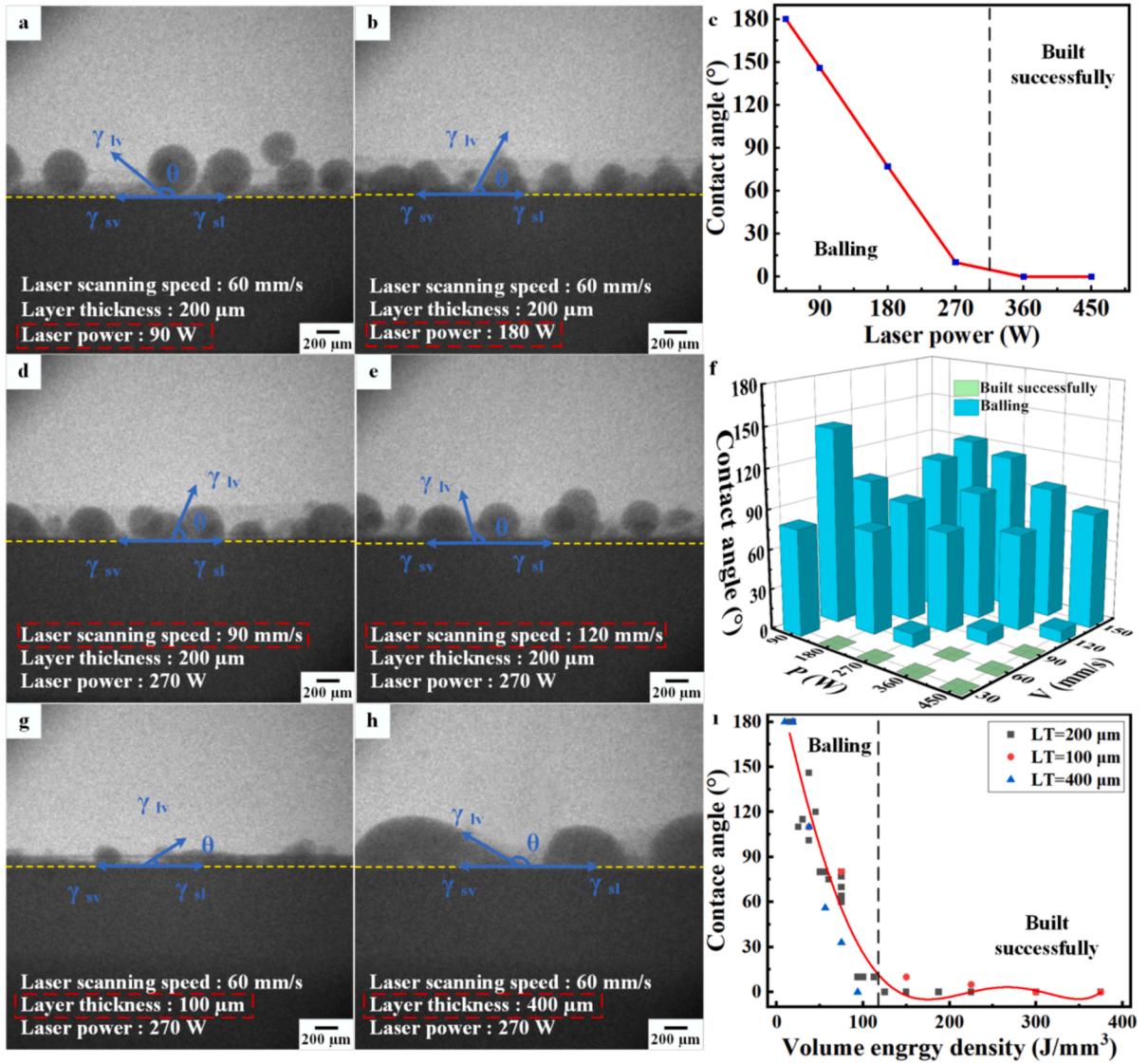


Fig. 11. Variation of the contact angle.

scanning speed the powder per unit volume absorbs less energy. The melt temperature is lower, leading to a shorter solidification time. On the contrary, high laser energy prolongs the solidification of melt. The statistical result shows the solidification times of the successfully built melt track (green part) and the balling (blue part) are between 20 ms–65 ms and 10 ms–20 ms, respectively.

3.3. Contact angle of the balls

The shape of molten metal depends on the gas–liquid–solid three-phase contact of the material, as shown in Fig. 10. It depends on the interfacial tension between the contact materials. The solid–liquid wettability is related to the solid–liquid interfacial tension (γ_{sl}), the solid–gas interfacial tension (γ_{sv}), and the liquid–gas interfacial tension (γ_{lv}), and the wettability can be characterized by the contact angle.

$$\cos\theta = \frac{\gamma_{sv} - \gamma_{sl}}{\gamma_{lv}} \quad (2)$$

The bad wettability between the molten metal and the surrounding metal powder particles and the substrate further promotes the formation of balling. For the given scanning speed of 60 mm/s (Fig. 11(a–c)), some powder remains on the substrate at the low laser power of 45 W, the melt

pool shape is mainly determined by the gas–liquid interfacial tension. The melt tends to agglomerate due to interfacial tension, leading to perfect balls with a contact angle of nearly 180°. When the powder is fully melted with the increase of laser power, the liquid–solid interfacial tension dominates the melt pool shape, and the contact angle gradually decreases. The contact angle is mainly between 137° and 150.7°, $\gamma_{sv} < \gamma_{sl}$, $\cos\theta < 0$, and the wettability of the liquid relative to the solid is poor at $P = 90$ W, which is prone to regular balls. Further increasing the laser power to 180 W, the improved wettability causes a smaller contact angle, forming the irregular balls. The liquid phase can better wet the solid phase at $P \geq 270$ W, and the contact angle is close to 0° [50]. Similarly, with the increase of the laser scanning speed or the lay thickness, the wettability becomes worse, and the contact angle gradually increases, as shown in Fig. 11(d–h).

The volumetric energy density was determined using Eq. (1). By integrating this calculation with experimental data, we utilized polynomial fitting techniques in Origin software to establish the fitting relationship (Eq. (3)) between the contact angle and the volumetric energy density.

The polynomial fitting equation for the balling formation at $Ev < 113.6 \text{ J}/\text{mm}^3$:

$$\theta = -1.63Ev + 185.16 (R_2 = 0.901) \quad (3)$$

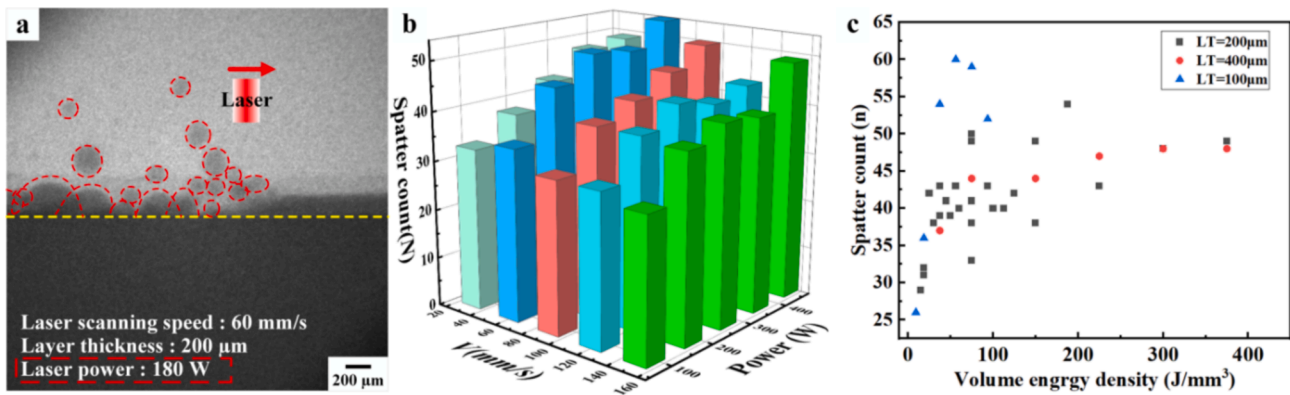


Fig. 12. Variation of the spatter count at high thick layers.

where θ is the Contact angle, $^\circ$; E_v is the volume energy density, J/mm^3 .

Analysis of the fitting data reveals (Fig. 11(i)) that when the volumetric energy density (E_v) is below $113.6 J/mm^3$, Due to insufficient laser energy, the balling phenomenon is pronounced. The threshold of volume energy density is about $113.6 J/mm^3$ for the balling formation. The contact angle is linear and negatively correlated with the volume energy density at $E_v < 113.6 J/mm^3$.

Eq. (2) is derived based on three layer thicknesses ($LT = 100, 200, 400 \mu m$), which may introduce some degree of error in the threshold values. Moreover, there are many parameters that influence energy density. We investigated the influence of other key parameters on the balling phenomenon, while keeping the particle size distribution and spot diameter of commonly used industrial 3D printing powders constant. Furthermore, the applicability of this equation to larger layer thicknesses requires further investigation.

3.4. Spatter count

The spatter quantity in the laser powder bed fusion process for high thick layers is illustrated in Fig. 12a–c. When the laser scan speed remains constant, an increase in laser power results in a higher temperature and larger size of the melt pool. This leads to more metal powder being vaporized, which in turn produces more spatter. High-power lasers cause the melt pool to become unstable, resulting in increased ejection of molten metal. The rapid evaporation of molten metal under high-power laser conditions generates bubbles and gas pressure that pushes more metal droplets away [51].

When the laser scan speed increases, the heating time of 316L per unit time decreases, leading to a reduction in the temperature of the melt pool. Higher scan speeds typically reduce spatter, as the shorter time allows for increased stability of the melt pool, thus decreasing the formation of bubbles and spatter. However, if the scan speed is too high, it can lead to incomplete melting of the powder and increased spatter. Therefore, the speed needs to be within an optimal range to balance melt pool stability and spatter quantity.

Low energy density results in insufficient melt pool temperature, preventing complete powder melting. With a smaller and cooler melt pool, spatter is relatively minimal, but this affects the final product's density and mechanical properties. At moderate energy density, the melt pool achieves appropriate stability, allowing the powder to melt evenly without excessive spatter, resulting in better surface quality and product accuracy. High energy density causes the melt pool temperature to rise excessively, leading to instability and increased spatter. Spatter not only impacts print quality but also affects subsequent processing steps of the powder bed [52].

4. Conclusions

The melting processes of 316 L powder at high layer thickness were directly observed using the in-situ micro-focus X-ray imaging system. The complex formation mechanisms of the balling and the effects of the spatters were revealed. Specific conclusions can be drawn as follows.

- (1) In addition to the shrinkage and solidification of the broken melt pool, the spatter has an important effect on the large-size balling formation. The spatters coalesce into big ones during flight and solidification, acting main source for the large-size balls. The solidification time of the balls ranges from 10 ms to 20 ms, which is positively correlated with the laser volume energy density.
- (2) The spatter coalescence event is not an occasional occurrence. Spatter coalescence rate (R_c) is introduced to quantify the spatter coalescence event. The 316L spatter coalescence rate ranges from 42.42 % to 73.04 %, which is positively correlated with the laser power.
- (3) At the high powder bed layer thicknesses of 100 μm , 200 μm , and 400 μm , the balling phenomenon occurs when the volume energy density is lower than the threshold value of $113.6 J/mm^3$. The contact angle of the balls is negatively correlated with the volume energy density ($\theta = -1.63E_v + 185.16$, $R^2 = 0.901$). The variation of spatter quantity under different process parameters was analyzed.

CRediT authorship contribution statement

Ziao Yan: Writing – review & editing, Writing – original draft, Investigation, Data curation. **Shicheng Liu:** Validation, Investigation. **Zhanpeng Sun:** Supervision, Project administration, Methodology, Conceptualization. **Kangshuo Li:** Investigation. **Nan Su:** Investigation. **Guang Yang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.matdes.2024.113442>.

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Data availability

Data will be made available on request.

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